UNIT – IV

Introduction to Ergonomics • Physical • Cognitive • Organizational • Environmental

Syllabus (as given)

Introduction to Ergonomics: human factors and ergonomics, psychology, engineering, biomechanics, industrial design, graphic design, statistics, operations research and anthropometry. Morphology of design and its relationship with cognitive abilities of human beings. Physical ergonomics: human anatomy, anthropometric, physiological and biomechanical characteristics as they relate to physical activity. Cognitive ergonomics: mental processes – perception, memory, reasoning, motor response, mental workload, decision-making. Organizational ergonomics: optimisation of socio-technical systems, organisational structures, policies, processes; communication, work design, design of working times, teamwork, cooperative work, new work programs. Environmental ergonomics: human interaction with the environment – climate, temperature, pressure, vibration, light.

1. Ergonomics – Definition, Origin and Objectives

Definition (IEA – International Ergonomics Association, 2000)

Ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data and methods to design in order to optimise human well-being and overall system performance.

- Word origin: Greek **ergon = work** + **nomos = natural laws** → "laws of work". Coined by K.F.H. **Murrell** (UK, 1949); the US term is **Human Factors (Engineering).** Other names: human engineering, biotechnology, engineering psychology.
- Simple definition: **fitting the job (task, tool, environment) to the person** – not the person to the job; "design for human use".

Brief history

- Early roots: Taylor's scientific management and the Gilbreths' motion study (work physiology, fatigue); Hawthorne studies (1924–32, social factors).

- World War II: aircraft cockpits, radar displays and weapons – operator errors due to poor design → **human factors engineering.** Post-war: Ergonomics Research Society (1949, UK); Human Factors Society (1957, USA); IEA (1959).
- Today: product design, computers & mobile devices, healthcare, transportation, safety, manufacturing, sports.

Objectives and benefits

- **Improve safety, health and comfort** – reduce accidents, musculoskeletal disorders (MSDs), fatigue and stress.
- **Improve performance & productivity** – reduce errors, increase speed and quality; reduce absenteeism, training time and cost.
- Improve user satisfaction, acceptability and usability of products and systems; match demands of the task with capabilities and limitations of humans.

2. Human Factors and Ergonomics – a Multidisciplinary Subject

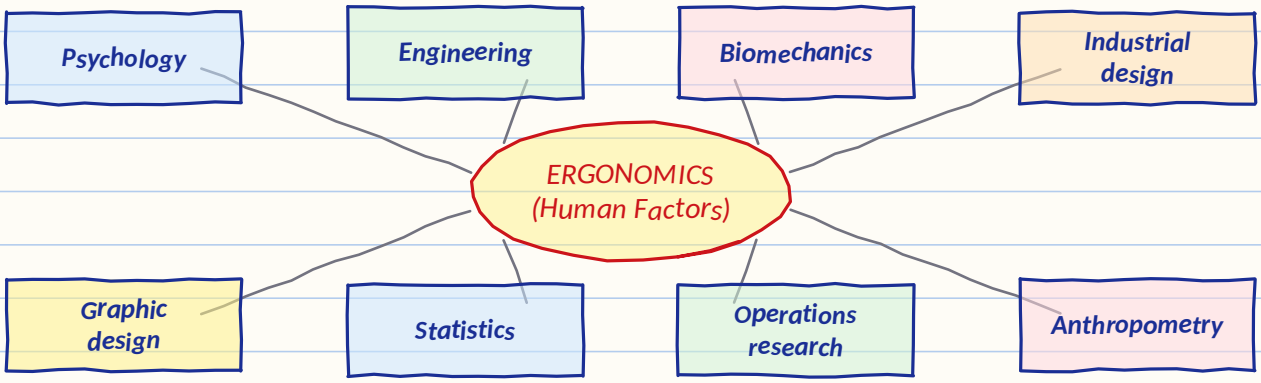


Fig 4.1 Ergonomics draws on many disciplines

Discipline	Contribution to ergonomics
Psychology	Perception, attention, memory, learning, motivation, stress, human error, mental workload, user behaviour
Engineering	Design of machines, controls, work systems, safety, human–machine system analysis, production & maintenance
Biomechanics	Application of mechanics to body: forces on joints & spine, lifting, posture, muscle strength, tool handles
Industrial design	Form, shape, aesthetics and usability of products, tools, furniture; human-centred styling
Graphic design	Visual communication: symbols, layout, colour, typography, displays, labels, user interface

Discipline	Contribution to ergonomics
Statistics	Design of experiments, sampling, percentiles, variability of human data, hypothesis testing
Operations research	Modelling & optimisation of human–machine systems, scheduling shift work, queuing, decision analysis
Anthropometry	Measurement of body dimensions; data for sizes of workspace, seats, tools, clothing; percentiles

- Also: physiology (work capacity, fatigue), anatomy, medicine / occupational health, sociology, architecture, computer science / HCI.

3. Morphology of Design and its Relationship with Cognitive Abilities

- **Morphology** = the study of form and structure. In design, **design morphology** means the systematic study of the form, structure, parts and their arrangement in a product (shape, size, layout, colour, texture, proportion) and how they are generated and combined.
- **Morphological analysis** (Zwicky): break the product into essential functions / parameters, list the possible options for each (a morphological matrix) and combine options to get new design concepts.

Function / parameter	Option 1	Option 2	Option 3
Grip of a hand tool	Cylindrical	Pistol grip	Contoured
Material	Wood	Plastic	Rubber coated
Activation	Push button	Trigger	Lever

- Stages of the design process: (1) problem / user analysis → (2) concept generation (morphology) → (3) embodiment: form, layout, dimensions → (4) detail design & prototype → (5) user testing & evaluation → iterate.

Relationship with cognitive abilities

- A product's form is "read" by the user's mind. If the form matches how humans **perceive, remember and think,** the product is easy to use; if not, errors and frustration occur.

Cognitive ability	Design implication (form / morphology)
Perception (Gestalt: proximity, similarity, closure, continuity)	Group related controls; use similar shape for similar function; clear visual hierarchy

Cognitive ability	Design implication (form / morphology)
Attention (selective, limited)	Make critical information salient: colour, size, position; avoid clutter
Memory (STM 7 ± 2 items; LTM)	Limit choices; chunking; recognition rather than recall; use familiar symbols
Mental models & expectations	Follow population stereotypes (clockwise = increase; red = stop)
Learning	Consistency, feedback, progressive disclosure, intuitive forms
Decision-making & reasoning	Present clear options, show state of system, support error recovery

Norman's design principles

- **Affordance** – the form suggests how it is to be used (a handle affords pulling).
- **Signifiers & visibility** – make possible actions visible.
- **Mapping** – natural spatial relation between controls and effects.
- **Feedback** – immediate indication of the result of an action.
- **Constraints** – form that prevents wrong action (poka-yoke); **consistency** across the system.

- Laws used in design: **Hick's law** – decision (choice reaction) time increases with log of number of choices: $RT = a + b \log_2(n + 1)$. **Fitts' law** – movement time to a target: $MT = a + b \log_2(2D / W)$ (D = distance, W = target width) – large, near targets are quicker.

4. Domains of Ergonomics (IEA)

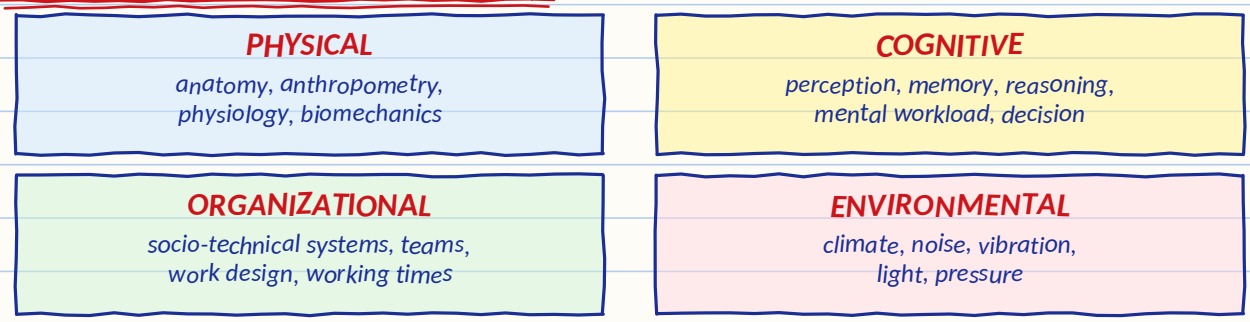


Fig 4.2 Four domains of ergonomics (IEA)

5. Physical Ergonomics

- Concerned with human anatomical, anthropometric, physiological and biomechanical characteristics as they relate to physical activity. Topics: working postures, materials handling, repetitive movements, work-related musculoskeletal disorders (WMSDs), workplace layout, safety and health.

(a) Human anatomy relevant to ergonomics

- **Skeletal system:** 206 bones – support, protect organs, levers for movement; **joints** (hinge – elbow, knee; ball & socket – shoulder, hip; pivot, gliding). The **spine** has 33 vertebrae: cervical (7), thoracic (12), lumbar (5), sacral (5 fused), coccyx (4 fused). Lumbar discs carry the highest load in lifting (L5/S1).
- **Muscular system:** skeletal muscle (voluntary), cardiac and smooth (involuntary). Muscle contraction: isotonic (movement), isometric (static holding – causes fatigue quickly), concentric / eccentric.
- Tendons connect muscle to bone; ligaments connect bone to bone; **nervous system** (CNS, PNS) controls; **circulatory & respiratory systems** supply O2 & remove waste (determine work capacity).
- Static muscular work restricts blood flow – leads to quick fatigue; dynamic work with rhythmic contraction is better.

(b) Anthropometry – body dimensions

- **Anthropometry** = measurement of the human body (size, shape, strength, mobility). Used to decide **dimensions of workplace, seats, controls, tools, clothing.**
- Types: **Static (structural)** – measured in fixed positions (stature, sitting height, arm length); **Dynamic (functional)** – measured during movement (reach envelope, range of joint motion); also strength measurement.

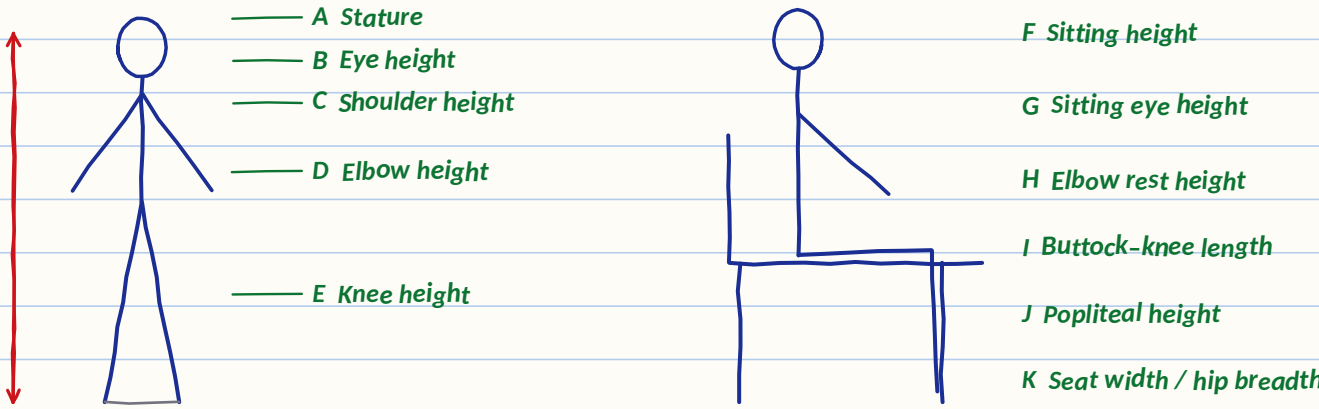
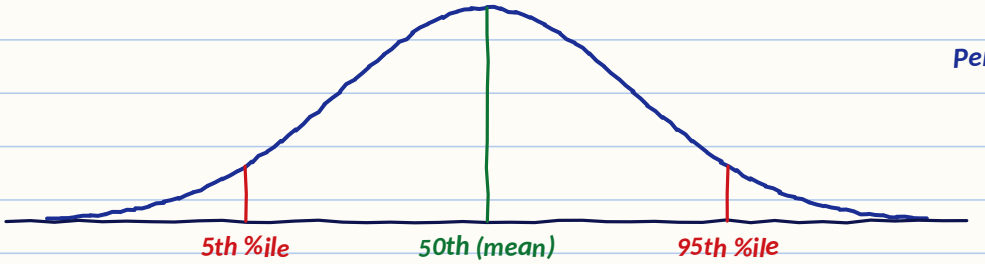


Fig 4.3 Common anthropometric measurements (standing & seated)

- Human data are **variable** – age, sex, race / region, occupation, generation (secular growth). Data are given as **mean, standard deviation and percentiles** (normal distribution).



Percentile value = mean ± z·σ
5th = mean – 1.645 σ
95th = mean + 1.645 σ
99th = mean + 2.33 σ

Fig 4.4 Normal distribution of a body dimension – percentiles

Principles of applying anthropometric data

Design principle	Use	Examples
Design for extremes (limiting dimension)	Min. or max. of population	Door height, clearance, leg room → 95th (max); reach to controls, seat height, force to operate lever → 5th (min)
Design for adjustability	Range 5th – 95th %ile (90 % of users)	Chair height, monitor height, car seat, steering
Design for the average	Where adjustment is not feasible	Counter height in a shop, hand-rails, checkout
Design for the individual	When dimension is for one person	Custom tool handle, prosthetics

Note: Always add allowances for clothing, shoes, posture slump, and use dimensions of the target population (e.g. Indian data).

(c) Physiological characteristics

- **Work physiology:** energy (ATP) supplied by aerobic and anaerobic processes; oxygen consumption (litres/min) and heart rate are measures of workload; 1 litre O_2 ≈ 5 kcal.

Work category	Energy expenditure (kcal / min)	Heart rate (beats / min)
Light	2.0 – 4.9	< 100
Moderate	5.0 – 7.4	100 – 120
Heavy	7.5 – 9.9	120 – 140
Very heavy	10.0 – 12.4	140 – 160
Extremely heavy	≥ 12.5	> 160

- Resting metabolism ≈ 1.5 kcal / min; recommended average workload for an 8-h shift ≈ 5.0 kcal/min (men) and ≈ 4.0 kcal/min (women) – above that rest pauses are needed.

Rest allowance – Murrell's formula

R = T × (W – S) / (W – 1.5) R = rest time (min), T = total work time (min), W = energy expenditure of the work (kcal/min), S = recommended sustainable level (5.0 men / 4.0 women).

Example: W = 7.5 kcal/min, T = 480 min (men): R = 480 × (7.5 – 5.0)/(7.5 – 1.5) = 480 × 2.5 / 6 = **200 min** of rest per shift.

- Fatigue: physical (muscle) and mental; causes – heavy / static load, long hours, monotonous work, poor environment; symptoms – reduced output, errors, discomfort. Remedy – job design, rest pauses, rotation, tools, training.
- Max. aerobic capacity (VO2 max) ≈ 3.0 l/min (men) and 2.0 l/min (women); the safe continuous workload ≈ 33 % of VO2 max for 8 h.

(d) Biomechanical characteristics

- **Biomechanics** = application of mechanics to the body: the body is treated as a system of **levers** (bones), **joints** (fulcrum) and muscles (effort). Important – forces, moments, posture, strength, tolerance of tissues.
- Principles: keep loads close to the body (short moment arm); avoid twisting & bending of the trunk; use legs not back; neutral posture of the joints; avoid static load; use mechanical aids; reduce repetition & force; reduce vibration.
- **WMSDs / RSI:** carpal tunnel syndrome, tendinitis, tennis elbow, low-back pain, neck & shoulder strain – risk factors: force, repetition, awkward posture, static load, duration, vibration, cold.
- Posture assessment tools: **OWAS, RULA** (Rapid Upper Limb Assessment), **REBA** (Rapid Entire Body Assessment); lifting tool: **NIOSH Lifting Equation.**

NIOSH lifting equation (metric)

RWL = LC × HM × VM × DM × AM × FM × CM (Recommended Weight Limit, kg)

LC = 23 kg (load constant); HM = 25/H (H = horizontal distance, cm); VM = 1 – 0.003 | V – 75|; DM = 0.82 + 4.5/D; AM = 1 – 0.0032 A (asymmetry); FM = frequency, CM = coupling.

Lifting Index LI = Load / RWL; LI > 1 → risk of injury – redesign task.

Example: H = 40, V = 60, D = 40, A = 0, FM = CM = 1 → HM = 0.625, VM = 0.955, DM = 0.9325 → RWL = 23 × 0.625 × 0.955 × 0.9325 ≈ **12.8 kg;** for a 15-kg load LI = 15/12.8 = 1.17 > 1.

6. Cognitive Ergonomics

- Concerned with mental processes – **perception, memory, reasoning, motor response** – as they affect interactions among humans and other elements of a system. Topics: mental workload, decision-making, skilled performance, human–computer interaction, human reliability, work stress and training.

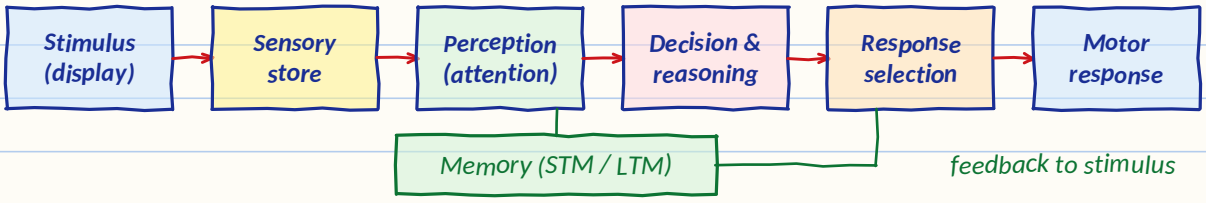
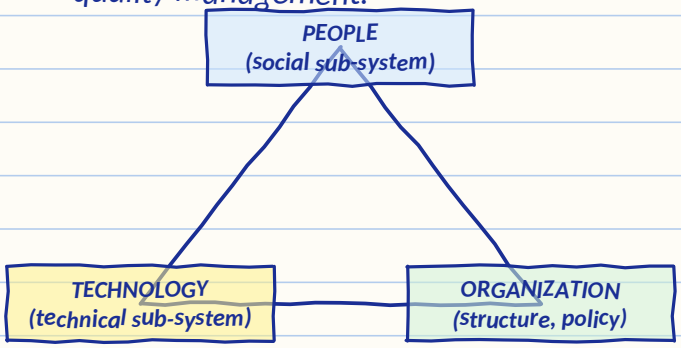


Fig 4.5 Human information-processing model

- **Perception:** sensing (vision, hearing, touch) and interpretation of stimuli; depends on contrast, size, expectation; illusions & limitations (absolute / difference thresholds – Weber's law $\Delta I / I$ = constant).
- **Attention:** selective, divided, sustained (vigilance). Vigilance falls after ≈ 30 min on monitoring tasks.
- **Memory:** sensory memory (≈ 1–4 s), short-term / working memory (≈ 7 ± 2 chunks, 15–30 s without rehearsal), long-term memory (large, permanent). Design: reduce memory load, use recognition.
- **Reasoning & decision-making:** deductive / inductive, heuristics and biases; skill-, rule- and knowledge-based behaviour (Rasmussen). Time pressure, uncertainty and information overload degrade decisions.
- **Motor response:** reaction time (simple visual ≈ 0.2–0.25 s, auditory ≈ 0.15–0.20 s, choice RT longer – Hick's law); speed–accuracy trade-off (Fitts).
- **Mental workload:** the demand placed on the human information-processing capacity. Measures: (i) primary / secondary task performance, (ii) **subjective** – NASA-TLX, SWAT, Cooper–Harper; (iii) **physiological** – heart rate variability, EEG, pupil diameter. Both under-load and overload are harmful.
- **Human error** (Reason): **slips** & **lapses** (right intention, wrong execution / memory failure), **mistakes** (wrong plan – rule / knowledge based), **violations.** Prevent via design: constraints, feedback, redundancy, checklists, training.
- **Situation awareness** (Endsley): perception of elements → comprehension → projection of future states. Important for pilots, control-room operators, drivers.

7. Organizational Ergonomics (Macro-ergonomics)

- Optimisation of **socio-technical systems** – including their organisational structures, policies and processes. Topics: communication, crew resource management, work design, design of working times, teamwork, participatory design, community ergonomics, cooperative work, new work paradigms, virtual organisations, telework, quality management.



Joint optimisation:
neither the technical nor the social part is optimised alone – both together (Tavistock studies – Trist & Bamforth)

Fig 4.6 Socio-technical system

Topic	Key points
Communication	Formal / informal, vertical & horizontal; clarity, feedback, handover (shift change), briefing, signs; avoids errors and accidents
Work design	Job enlargement (more tasks), job enrichment (more responsibility), job rotation (reduce monotony), autonomous work groups, ergonomic task allocation (human vs machine)
Design of working times	Hours of work, rest pauses, shift systems (forward rotation better), night work & circadian rhythm (body temperature lowest ≈ 3–4 am), flexi-time, compressed work week, overtime limits
Teamwork & cooperative work	Clear roles, shared goals, trust; groupware (CSCW), team training, CRM (aviation); social loafing risk
New work programs	Telework / work-from-home, hot-desking, virtual teams, lean production, TQM, participatory ergonomics, safety culture, gig / platform work
Policies & processes	Health & safety policy, incentive schemes, training, staffing levels, management style, participation of workers

Note: Participatory ergonomics: involve workers in identifying and solving ergonomic problems – high acceptance of changes.

8. Environmental Ergonomics

- Concerned with human interaction with the physical environment characterised by **climate (temperature, humidity, air movement), pressure, vibration, noise and light** – and how they affect comfort, health, safety and performance.

(a) Climate / thermal environment

- Heat balance: $M \pm R \pm C - E = S$ (metabolic heat, radiation, convection, evaporation, storage). Four factors: air temperature, humidity, air velocity, mean radiant temperature (+ clothing & activity).
- Thermal comfort zone (sedentary): air temp ≈ 20–26 °C, RH 40–60 %, air speed < 0.2 m/s. Indices: **Effective Temperature (ET), WBGT,** PMV – PPD (Fanger; ISO 7730).
- **WBGT** = 0.7 Tnwb + 0.3 Tg (indoors / no sun); = 0.7 Tnwb + 0.2 Tg + 0.1 Ta (outdoors with sun). Higher WBGT → shorter work period / more rest.
- Heat stress → heat cramps, exhaustion, **heat stroke;** cold stress → hypothermia, frostbite, reduced dexterity (hands < 15 °C). Control – engineering (ventilation, insulation, shields), administrative (work–rest, acclimatisation), PPE.

(b) Pressure

- High altitude (hypobaric): low O2 partial pressure → hypoxia, reduced performance; need acclimatisation / oxygen. Hyperbaric (diving, tunnelling, caissons): decompression sickness ("bends"), nitrogen narcosis – staged decompression is necessary.

(c) Vibration

- **Whole-body vibration** (vehicles, cranes; 0.5–80 Hz): spine resonates at ≈ 4–8 Hz, discomfort, back pain, motion sickness (0.1–0.5 Hz). Standard: ISO 2631.
- **Hand-arm vibration** (power tools; 8–1000 Hz): vibration white finger (Raynaud's), carpal tunnel; ISO 5349 – limit exposure A(8) 5 m/s^2.
- Control: isolation (springs, dampers), anti-vibration handles, seat suspension, reduce exposure time, maintenance, gloves.

(d) Noise

- Sound level measured in dB(A). Permissible exposure ≈ 85 dB(A) for 8 h (NIOSH, with 3-dB exchange rate: halve time for each +3 dB); 90 dB(A) (OSHA / Indian factories norm, 5-dB exchange). Effects: hearing loss (NIHL), annoyance, communication interference, stress, errors.
- Control: at source (quiet machines, damping), path (enclosures, barriers, absorbers), receiver (ear plugs / muffs, job rotation); hearing conservation programme (audiometry).

(e) Illumination / light

- Factors: illuminance (lux), luminance, contrast, glare (direct & reflected), colour, uniformity, age of worker. Poor light → eye strain, headache, errors, accidents.

Task / area	Typical illuminance (lux)
Corridors, stores, passages	50 – 100
Rough work, general factory, machine shops	200 – 300
Office work, reading, computer area	300 – 500
Drawing office, medium assembly, inspection	500 – 750
Fine assembly, instrument work, colour matching	750 – 1000
Very fine work (watch making, micro-electronics)	1500 – 2000 + local light

- Design: use task lighting + general lighting; avoid glare (shielding, matt surfaces, correct angle), ratio of brightness between task : surroundings ≤ 3 : 1 (near), ≤ 10 : 1 (far); colour of light (Ra > 80); maintain lamps.
- Other environmental factors: air quality / ventilation, dust & fumes, humidity, radiation, colour & aesthetics, room layout, hygiene, ergonomic chair / workstation.

9. Application – Ergonomic Workstation Design (Quick Guide)

Item	Guideline
Seated work-surface height	About elbow height when seated (≈ 70–75 cm for keyboard work); adjustable table preferred
Chair	Seat height adjustable (≈ 38–50 cm, popliteal height); seat depth 38–43 cm; lumbar support; backrest 95–110°; castors, arm rests; feet flat on floor / footrest
Monitor / VDT	Top of screen at or slightly below eye level; viewing distance 50–70 cm; line of sight 15–20° below horizontal; no glare; text size ≥ 20 minutes of arc
Keyboard & mouse	Elbows ≈ 90°; wrists neutral (not bent); keyboard at elbow height; mouse close; palm support
Standing work	Precision work: surface 5–10 cm above elbow; light assembly: 5–10 cm below elbow; heavy work: 15–40 cm below elbow height
Breaks	Micro-breaks every 30–45 min; 20-20-20 rule for eyes (every 20 min look 20 feet / 6 m away for 20 s); alternate sit–stand

- General rules: **neutral posture, elbows at sides,** keep frequently-used items within the normal reach area (35–40 cm), heavy items between knee and shoulder height, avoid static positions, vary tasks, provide adjustability and good lighting.
- Ergonomic evaluation process: identify task → collect data (observation, video, checklists, questionnaires, body-part discomfort map) → analyse risk (RULA / REBA / NIOSH) → redesign (engineering controls first, then administrative, PPE last) → implement → evaluate and follow-up.

Risk factor	Typical ergonomic control
Excessive force / lifting	Mechanical aids, lighter loads, team lifting, better handles & grip, adjustable height tables
Repetition	Automation, job rotation, rest pauses, power tools, task enlargement
Awkward / static posture	Adjustable workstation, re-position work, supports, tilting tables, sit-stand
Vibration / cold / contact stress	Anti-vibration tools, gloves, padding, rounded edges, warm environment

10. Quick Revision & Probable Questions

Memory hooks

- Ergonomics = ergon + nomos; fit the job to the person. Domains: Physical – Cognitive – Organizational – Environmental.
- Disciplines: Psychology, Engineering, Biomechanics, Industrial design, Graphic design, Statistics, OR, Anthropometry.
- Percentiles: 5th = mean – 1.645σ; 95th = mean + 1.645σ. Extremes: clearance → 95th; reach → 5th.
- NIOSH: $RWL = 23 \times HM \times VM \times DM \times AM \times FM \times CM$; LI = load / RWL.
- Murrell: $R = T\,(W - S)/(W - 1.5)$. $WBGT = 0.7\,Tnwb + 0.3\,Tg$.

1. Define ergonomics. Explain its history, objectives and the disciplines contributing to it.
2. Explain design morphology and its relationship with the cognitive abilities of human beings.
3. What is anthropometry? Explain the principles of applying anthropometric data with examples.

4. Explain physical ergonomics – anatomy, physiology and biomechanics related to physical activity.
5. Discuss cognitive ergonomics: perception, memory, reasoning, mental workload and decision-making.
6. What is organisational ergonomics? Discuss work design, working times and teamwork.
7. Explain environmental ergonomics – climate, noise, vibration, illumination and pressure.